

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of M. Pharm (Quality Assurance)

University Theory Examination January 2018

PHQA001 MODERN ANALYTICAL TECHNIQUES

Date: 01.01.2018 , Monday

Time: 10.00 a.m. to 01.00 p.m

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1 (A) Explain the basic elution principle for column and planar chromatography. 05
(B) Give classification of stationary phase used in Gas chromatography (GC) and explain the factors governing the selection of stationary phase in GC. 05
- Q.2 (A) Write a note on factors affecting separation in HPLC. 05
(B) Describe in brief mobile phase selection and optimization for HPTLC. 05
- OR
- Q.2 (A) Explain the fragmentation rules in mass spectroscopy. 05
(B) Discuss the principle of affinity and size exclusion chromatography. 05
- Q.3 Answer the following (Any three) 15
(A) Differentiate between DSC and DTA.
(B) What is MALDI? Explain its working in brief.
(C) Differentiate between GC and HPLC.
(D) Write a note on capillary electrophoresis.

SECTION-II

- Q.4 (A) Discuss different types of electronic transitions observed in spectroscopy. 05
(B) Explain different factors affecting absorption maxima and absorption spectra in UV-visible spectroscopy. 05
- Q.5 (A) Write note on Derivative spectroscopic method used for analysis of combination drugs. 05
(B) Discuss Michaleson Interferometer and sampling techniques used in IR spectroscopy. 05
- OR
- Q.5 What is Fluorescence spectroscopy? Discuss the instrumentation of fluorescence spectroscopy. Write factors affecting fluorescence spectra. 10
- Q.6 Answer the following (Any three) 15
(A) Write short note on: 2D NMR technique.
(B) What is X-ray diffraction technique? Discuss Bragg's law.
(C) Write note on ELISA.
(D) Write a note on ¹³C NMR technique.
